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MILITARY SPECIFICATION

WINCH, DRUM, POWER OPERATED

(TYPES 11.0-B-1D, 11.0-B&S.W.(AT)-2D

and 11.0-B&S.W.(AT)-2D/6.0-2G)

1. SCOPE

1.1 Scope. This specification covers the detail requirements for underway replenishment (UNREP) type 11.0-B-1D burton winches and 11.0-B&S.W.(AT)-2D and 11.0-B&S.W.(AT)-2D/6.0-2G burton and spanwire winches.

1.2 Classification. Winches shall be classified in accordance with the following types:

Type 11.0-B-1D - Single drum burton winch rated 11,000 pounds at 220 feet per minute.
Type 11.0-B&S.W.(AT)-2D - Double drum burton and spanwire winch with automatic tension. Winch rated 11,000 pounds at 220 feet per minute.
Type 11.0-B&S.W.(AT)-2D/6.0-2G - Double drum double gypsy burton and spanwire winch with automatic tension. Winch rated 11,000 pounds at 220 feet per minute and gypsy unit rated 6,000 pounds at 120 and 60 feet per minute.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
RR-W-410 - Wire Rope and Strand.

MILITARY
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-W-15802 - Winch, Gypsy, Power Operated, Electric.
MIL-B-16392 - Brakes, Magnet, Naval Shipboard.
MIL-M-17060 - Motors, 60 Cycle, Alternating-Current, Integral Horsepower (Shipboard Use).
MIL-M-17265 - Winch, Drum, Power Operated (Electric-Hydraulic, Variable Speed).
MIL-C-23742 - Control Centers, Alternating Current, Naval Shipboard.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Winch types 11.0-B-1D, 11.0-B&S.W.(AT)-2D and 11.0-B&S.W.(AT)-2D/6.0-2G shall be in accordance with MIL-W-17265, except as specified herein.

3.2 Construction.

3.2.1 Type 11.0-B-1D. Type 11.0-B-1D is a single drum single gypsy winch to be used basically for burtoning from a carrier.



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3.2.2 Type 11.0-B&S.W.(AT)-2D. Type 11.0-B&S.W.(AT)-2D is a double drum winch. One drum will be used basically for burtoning and the other drum will be used for handling the spanwire. The drum designated for the spanwire shall be provided with speed and tension modes of operation and the drum designated for burtoning shall be provided with only a speed mode of operation. This winch is intended basically to be used on a carrier. Only one drum will be used at a time.

3.2.3 Type 11.0-B&S.W.(AT)-2D/6.0-2G. Type 11.0-B&S.W.(AT)-2D/6.0-2G is a double gypsy winch. One of the drum units will be used basically for burtoning and the other drum will be used for handling the spanwire. The gypsy heads will be used basically for handling the saddles. The drum designated for the spanwire shall be provided with speed and tension modes of operation and the drum designated for burtoning shall be provided with only a speed mode of operation. This winch is intended basically to be used on a carrier. Only one of the drums will be used at a time. The gypsy heads can be used simultaneously with one of the drum units.

3.3 Drums. Drums shall be designed for the following:

(a) Stowage capacity for 800 feet of 3/4 inch wire rope. The breaking strength of the wire rope is 48,600 pounds.
(b) Barrel diameter:
    (1) Burton drum - 24 inches plus or minus 1/2 inch.
    (2) Spanwire drum - 32 inches plus or minus 1/2 inch.
(c) Distance between flanges shall be 15 inches plus or minus 1/2 inch.

3.4 Wire rope construction. Wire rope for the winch shall be type 1, class 3, construction 6, independent wire rope core (IWRC), improved plow steel, uncoated, preformed, right regular lay in accordance with RR-W-410.

3.5 Gypsy heads. Winch type 11.0-B-1D shall be provided with one gypsy head. (See 3.17.1 for gypsy head requirements for winch type 11.0-B&S.W.(AT)-2D/6.0-2G.)

3.6 Gearing. The gear drives provided shall have a service factor of 1.25.

3.7 Sheaves. All sheaves shall be designed with a minimum sheave tread diameter to wire rope diameter ratio of 24 to 1.

3.8 Torque clutch. Each drum shall be equipped with a torque (slip) clutch. This clutch shall be the driving member between the drum shaft and the drum. The tension (slip) value specified in 3.8.1 is based on the wire being on the average layer of the drum. The clutch shall not use any external energy source and shall be self compensating for wear. The slip value of the clutch shall not be effected by environments specified for the winch or clutch inactivity (no slippage) for the design life of the winch.

3.8.1 The torque clutch for the winch shall slip at a maximum of 22,500 pounds and reset at a minimum of 17,500 pounds.

3.9 Controlled spooling and anti-slack devices. Each winch drum shall be provided with controlled spooling and an anti-slack devices to insure proper spooling of wire on the drum. Proper spooling shall exist at all times the equipment is in operation as well as during start-up, after periods of disuse, without making adjustments. Proper spooling shall exist during periods of maximum winch deceleration and maximum acceleration between full speed haul-in and full speed payout for all winch loads from zero to rated load.

3.9.1 Controlled spooling. The controlled spooling device shall consist of a grooved drum, power operated level wind and a swiveling fairleader.

3.9.1.1 Grooved drum. The grooved drum shall be designed so that wire spools shall be provided on the drum in an evenly spaced, interrupted spiral pattern from flange to flange. The first and each succeeding layer of wire on the drum shall form a spooling pattern for the next layer that will exactly reproduce this spooling pattern on all layers. The spooling pattern of wire on the drum shall eliminate overriding (cross over) turns and undercutting of the wire rope and result in a minimum amount of scrubbing and chafing of the wire rope.

3.9.1.2 Level wind. A power operated level wind or similar device, automatically operated, shall be provided as a part of the spooling system.



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3.9.1.3 Swiveling fairleader. A swiveling fairleader shall be provided as a part of the spooling system. The fairleader shall be designed so that it can be readily locked in any operating position. The swiveling fairleader shall be mounted on the winch foundation.

3.9.2 Anti-slack device. The anti-slack device shall prevent slack wire from developing between the fairleader and the drum. The anti-slack device shall be hydraulically driven and shall normally be connected to the sheave of the swiveling fairleader (see 3.9.2.

3.9.2.1 ON-OFF switch. A snap action ON-OFF switch shall be provided on the control console to permit the operator to activate or deactivate the anti-slack device.

3.9.2.2 Selector switch. On the double drum winches a selector switch shall be provided on the console to permit the operator to select which anti-slack device will be activated. The selector switch may be combined with the ON-OFF switch.

3.9.2.3 Automatic cut-off. An automatic cut-off switch shall be provided to deactivate the anti-slack device when loads in excess of 1500 pounds are placed on the winch.

3.9.2.4 Deactivating the anti-slack device. When the anti-slack device is deactivated the pumping unit and the hydraulic motor shall be disengaged or otherwise protected from overspeeding.

3.9.2.5 Alternate installation. The anti-slack device shall be designed so that it can be removed from the swiveling fairleader and installed in front of the winch either on the deck or on the bulkhead. Installation instructions shall be included in the technical manual for both types of installation. The part number of the replacement sheave for the swiveling fairleader shall be provided.

3.9.3 Wire rope life requirements. With the anti-slack device energized, no load on the winch and the winch operated at rated speed, the anti-slack device and controlled spooling system shall provide at least 15,000 cycles of operation without failure (see 3.9.6) of the wire rope. With the anti-slack device turned off, rated load on the winch and the winch operated at rated speed, the anti-slack device and controlled spooling system shall provide at least 30,000 cycles of operation without failure of the wire rope.

3.9.4 Cycle. A cycle is defined as passage of a section of wire rope from the winch drum through the winch controlled spooling and anti-slack system and back onto the winch drum.

3.9.5 Cycle rate. Cycle rate for the wire rope life requirements specified in 3.9.3 is one cycle every 20 seconds.

3.9.6 Wire rope failure. Wire rope failure is defined as seven or more broken wires in one strand in one rope lay.

3.9.7 Wire rope lead. The wire rope shall lead from the drum to the level wind then to the line sensing device (applicable only where tension mode of operation is required) and then to the swiveling fairleader or anti-slack device. The swiveling fairleader and anti-slack device shall be designed so that the wire can be led from the swiveling fairleade or anti-slack at any angle up to 60 degrees each side of a vertical plane normal to the winc drum (see top view of figure 1) and simultaneously lead at any angle from 0 degrees to 75 degrees above the horizontal (see side view of figure 1).

3.10 Modes of operation. All winch types shall be provided with speed mode of operation. The drum used for handling the spanwire on winch types 11.0-B&S.W.(AT)-2D and 11.0-B&S.W.(AT)-2D/6.0-2G shall be provided with tension mode as well as speed mode of operation.

3.10.1 Speed control mode of operation. The rated line pull and rated line speed on the average layer for speed mode of operation shall be as follows:

(a) Burton winch drum - Rated line pull of 11,000 pounds on the average layer of the drum and a rated line speed of 220 feet per minute.
(b) Spanwire winch drum - Rated line pull of 8,500 pounds on the average layer of the drum and a rated line speed of 300 feet per minute.

3.10.2 Automatic tension mode of operation.

3.10.2.1 Line tension. Line tension shall be determined by a line sensing device.



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3.10.2.2 Performance.

3.10.2.2.1 The automatic tension control of the winch shall be designed so that it will cause the winch to payout or haul-in to maintain required tension level.

3.10.2.2.2 The speed-tension values at maximum tension setting shall follow the curve formed by points A, B, C, D, and E (see figure 2). The speed at point A is 540 feet per minute (payout) and point E the speed is 540 feet per minute (haul-in). Curve DE shall be a constant power curve formed by the following equation:

Speed (fpm haul-in) times tension (pounds) equals $25.5 \times 10^5$.

3.10.2.2.3 The curve from point A to B (see figure 2) shall be a straight line and shall have a tension increase of one pound for each foot per minute speed increase. The speed at point B is zero and the tension 9500 pounds.

3.10.2.2.4 The curve from point B to C (see figure 2) shall be a straight line. The tension at point B is 9500 pounds and the tension at point C is 8000 pounds.

3.10.2.2.5 The curve from point C to D (see figure 2) shall be a straight line and shall have a tension decrease of one pound for each foot per minute speed increase.

3.10.2.2.6 At reduced tension setting the speed-tension curve (see figure 2) shall parallel ABCD until the speed reaches 540 feet per minute or intersects curve DE in which case it will follow curve DE to point E.

3.10.2.2.7 At minimum tension setting point B' (see figure 2) shall be 3500 pounds and point C' shall be 2000 pounds.

3.10.2.2.8 The winch design shall permit the winch to be overhauled by an external force at speeds from 540 to 700 feet per minute. The tension value during this period may be any value required to protect winch components from damage including power generation, but shall not exceed 10,200 pounds.

3.10.2.2.9 The winch design shall provide automatic recovery from the 540 to 700 feet per minute overhauling speed condition back to normal tension control range (540 feet per minute payout to 540 feet per minute haul-in).

3.10.2.2.10 The design of the automatic tension control shall be such that reference points A, B, C and D (see figure 2) could be modified either through adjustments or substitution of components.

3.10.2.2.11 Actual tension values shall be within plus or minus 200 pounds of the values specified in 3.10.2.2.2 through 3.10.2.2.8.

3.10.2.2.12 The winch shall provide tension accuracy at all tension settings of plus or minus 15 percent of the rated tension during acceleration and deceleration at rates up to 6.7 feet per second squared.

3.10.2.2.13 The rate at which the tension level can be manually raised from minimum to maximum values shall be restricted to a minimum of three seconds.

3.11 Duty. Winch type 11.0-B-1D and the drum unit of winch types 11.0-B&S.W.(AT)-2D and 11.0-B&S.W.(AT)-2D/6.0-2G shall be designed to permit continuous operation for a period of six hours. A normal operating cycle will consist of 10 seconds at rated speed and rated load; 10 seconds at idle; 10 seconds at half-speed and rated load; and 10 seconds at idle. (See 3.17.2 for duty requirements for gypsy unit of winch type 11.0-B&S.W.(AT)-2D/6.0-2G.)

3.11.1 Nameplate ampere rating of the electric motor shall not be exceeded when the winch is being operated at rated load and rated speed (constant velocity), temperature rise shall not exceed nameplate permissible temperature rise when the winch is operated at the duty cycle and period specified in 3.11.

3.11.2 Thermal overload shall be indicated by energizing a master panel indicating light through a thermal heating device (see thermal requirements of MIL-W-17265) in either the motor or the controller. The indicator light shall be in accordance with the illumination requirements of MIL-W-17265. This device shall not de-energize the motor.



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3.12 Hydraulic brake. When in speed control mode of operation the hydraulic brake shall automatically set upon loss of servo pressure, replenishment pressure or electric power. When in automatic tension mode of operation the hydraulic brake shall be designed to stay in the released position for at least 15 minutes upon loss of servo pressure, replenishment pressure or electric power.

3.13 Relief valve. The high pressure relief valve (hoist side) for the hydraulic system on winch types 11.0-B&S.W.(AT)-2D and 11.0-B&S.W.(AT)-2D/6.0-2G shall automatically vent to a lower pressure value to limit the maximum load that can be imposed on the wire rope. The high pressure relief valve shall only vent to a lower value when the winch is in tension mode and the winch control goes into fail safe or electric power to the winch is shut off. The high pressure shall remain at its normal setting whenever the winch is in speed mode or when the winch is operated from the emergency mechanical speed control unit.

3.13.1 The lower pressure for the relief valve shall be adjustable and shall be set at a value that will permit the winch to be overhauled with a rope pull of 4500 to 5500 pounds at speeds from zero to 540 feet per minute (wire rope on the average layer) when operating in tension mode of operation. This requirement is based on the hydraulic brake being released (see 3.12).

3.14 Hydraulic motors. The hydraulic motor may be of the variable delivery type. In the event of loss of power to the system the hydraulic motor, if variable delivery type, shall go to minimum stroke.

3.15 Methods of operation. Master speed control and winch speed control units confor ing to the requirements of MIL-W-17265 shall be provided. In addition a master tension control unit conforming to 3.15.1 shall be provided for winch types 11.0-B&S.W.(AT)-2D and 11.0-B&S.W.(AT)-2D/6.0-2G.

3.15.1 Master tension control. A master tension control shall be added to the operations selector switch (see MIL-W-17265 for requirements for master control panel).

3.15.1.1 Associated controls required to perform the functions of 3.10.2 shall also be added to the master control panel.

3.15.1.2 The lever for the master speed function shall control the tension operation as follows:

(a) When the operations selector switch is turned counterclockwise to the master tension position, the lever arm, in its spring centered position shall establish minimum tension in the system.
(b) Pulling the lever arm towards the operator shall increase tension to maximum
(c) Returning the lever arm forward to the center position shall decrease tensio to minimum.
(d) The system shall remain in minimum tension for all lever positions forward of center position.
(e) For tension operation it shall be possible to readily lock the control handl in any given position. The locking device shall be designed so that the operator may override it if necessary to change the tension.

3.15.2 Replenishment-at-sea control stations. A replenishment-at-sea control station shall be provided for each winch. Each control station shall include all controls and indicating lights specified for the various winch types. The controls shall be arranged on an integrated console.

3.16 Size and weight. The winch shall be as compact and light as is consistent with adequate strength and rigidity. Weight and size of the various types shall not exceed the following:

(a) Type 11.0-B-1D:
  (1) Maximum weight - 15,500 pounds.
  (2) Maximum width (across drum) with gypsy head installed - 90 inches.
  (3) Maximum length (perpendicular to drum) - 140 inches.
  (4) Maximum height - 90 inches.
(b) Type 11.0-B&S.W.(AT)-2D:
  (1) Maximum weight - 17,300 pounds.
  (2) Maximum width (across drums) - 80 inches.
  (3) Maximum length (perpendicular to drum) - 150 inches.
  (4) Maximum height - 90 inches.



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(c) Type 11.0-B&S.W.(AT)-2D/6.0-2G:
(1) Maximum weight - 20,000 pounds.
(2) Maximum width (across drums) with gypsy heads installed - 114 inches.
(3) Maximum length (perpendicular to drum) - 150 inches.
(4) Maximum height - 90 inches.

3.17 Gypsy winch. The gypsy winch unit of winch type 11.0-B&S.W.(AT)-2D/6.0-2G shall be in accordance with MIL-W-15802, except as specified herein. Requirements of MIL-W-15802 that conflict with those of MIL-W-17265, the requirements of MIL-W-17265 shall govern.

3.17.1 Gypsy heads. Gypsy heads shall be proportioned for five inch circumference manila rope, maximum breaking strength of 22,500 pounds.

3.17.2 Gypsy winch electric motor. The gypsy winch electric motor shall be in accordance with MIL-W-17060 and the following:

| | |
|---|---|
| Classification | Navy A |
| Ambient temperature | 40°C. |
| Voltage | 440 a.c. |
| Phase | 3 |
| Cycles per second | 60 |
| Duty | Full load at half speed, 1/2 hour; 1/2 load at half speed, continuous |
| Enclosure | Watertight |
| Speed | Two speed (full and half) constant torque |
| Type | Squirrel cage induction |
| Design | D |
| Horsepower | As required |

3.17.3 Gypsy winch motor controller. The gypsy winch electric motor controller and master switch shall be in accordance with MIL-C-2212 or MIL-C-23742 as applicable and the following:

| | |
|---|---|
| Operation | Magnetic |
| Voltage | 440 v.a.c. 3 phase, 60 cycles |
| Type | Across the line |
| Performance | Nonautomatic |
| Function | Motor starting, reversing and speed selection |
| Protection | Low voltage release and thermal overload |
| Enclosure | Dripproof protected |
| Duty | Continuous |
| Emergency run feature | Remote emergency run pushbutton shall be provided with a watertight housing |
| Ambient temperature | 50°C. |
| Master switch | Remote drum type, lever operated spring return to OFF, watertight enclosure with power ON indicating light |

3.17.4 Gypsy winch electric brake. The gypsy winch electric brake shall be in accordance with MIL-B-16392 and the following:

| | |
|---|---|
| Voltage | 440 a.c. |
| Duty | Continuous |
| Ambient temperature | 40°C. |
| Enclosure | Watertight |
| Mounting | As required |

3.17.5 Gypsy winch rated line pull and line speed. The rated line pull and rated line speed of the winch shall be as follows:

(a) Rated line pull - 6000 pounds.
(b) Rated line speed - 120 and 60 feet per minute.

4. QUALITY ASSURANCE PROVISIONS

4.1 In addition to the tests specified in MIL-W-17265, the tests specified in 4.2 through 4.5 shall be conducted for winch types 11.0-B-1D and 11.0-B&S.W.(AT)-2D and the drum unit of winch type 11.0-B&S.W.(AT)-2D/6.0-2G.



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4.2 Controlled spooling system and wire rope life test. One winch of each type offered for delivery on one contract or order shall be tested to show conformance with all requirements specified for the controlled spooling and anti-slack device system and for wire rope life (see 3.9 through 3.9.6).

4.3 Torque clutch test. Torque clutch tests shall be made at the manufacturer's plant.

4.3.1 Each drum torque clutch shall be tested by pulling a 40 foot length or wire off the drum at a speed of 200 feet per minute. The tensions measured during this test shall be recorded. (The use of a special locking device on the winch to hold the drum shaft stationary during this test is acceptable.)

4.3.2 The torque clutch of one winch of each type in the lot shall be tested by repeating the above test two times, waiting approximately 15 seconds between runs, and seven additional times, allowing the clutch to cool as necessary between runs. The maximum tension measured during each run shall be recorded.

4.4 Vented relief valve test. Each winch shall be tested to demonstrate that the vented relief valve has been properly set.

4.5 Tension tests. Each winch shall be tested to show conformance with tension mode of operation (see 3.10.2 through 3.10.2.2.13).

4.6 Gypsy winch tests. The gypsy winch unit of winch type 11.0-B&S.W.(AT)-2D/6.0-2G shall be tested in accordance with the requirements of MIL-W-15802.

4.7 Procedures. The procedures for conducting tests specified in 4.2 through 4.5 shall be submitted to the Naval Ship Engineering Center (NAVSEC) Weapons and Deck Equipment Branch, Department of the Navy, Center Building, Prince George's Center, Hyattsville, Maryland 20782.

4.8 Record of tests. A copy of data recorded in conducting the tests specified in 4.2 through 4.5 shall be forwarded to NAVSEC within thirty days after completion of tests.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-W-17265.

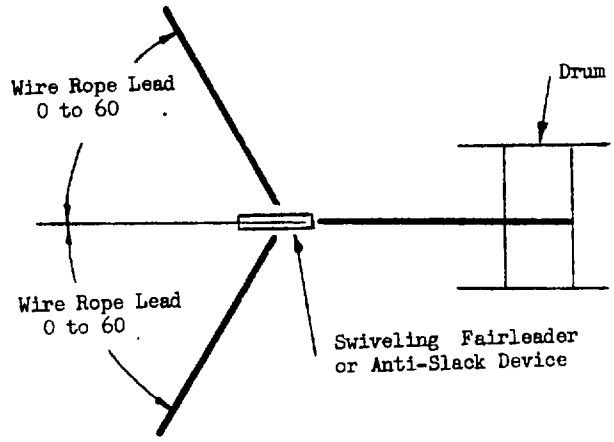
6. NOTES

6.1 Ordering data. Ordering data shall be as specified in section 6 of MIL-W-17265.

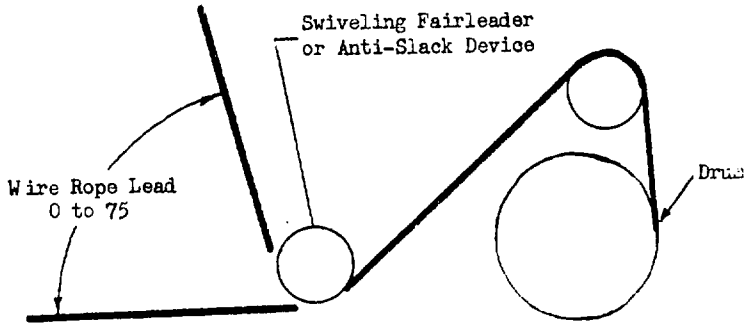
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TOP VIEW




SIDE VIEW

Figure 1 - Wire rope lead from swiveling fairleader or anti-slack device.



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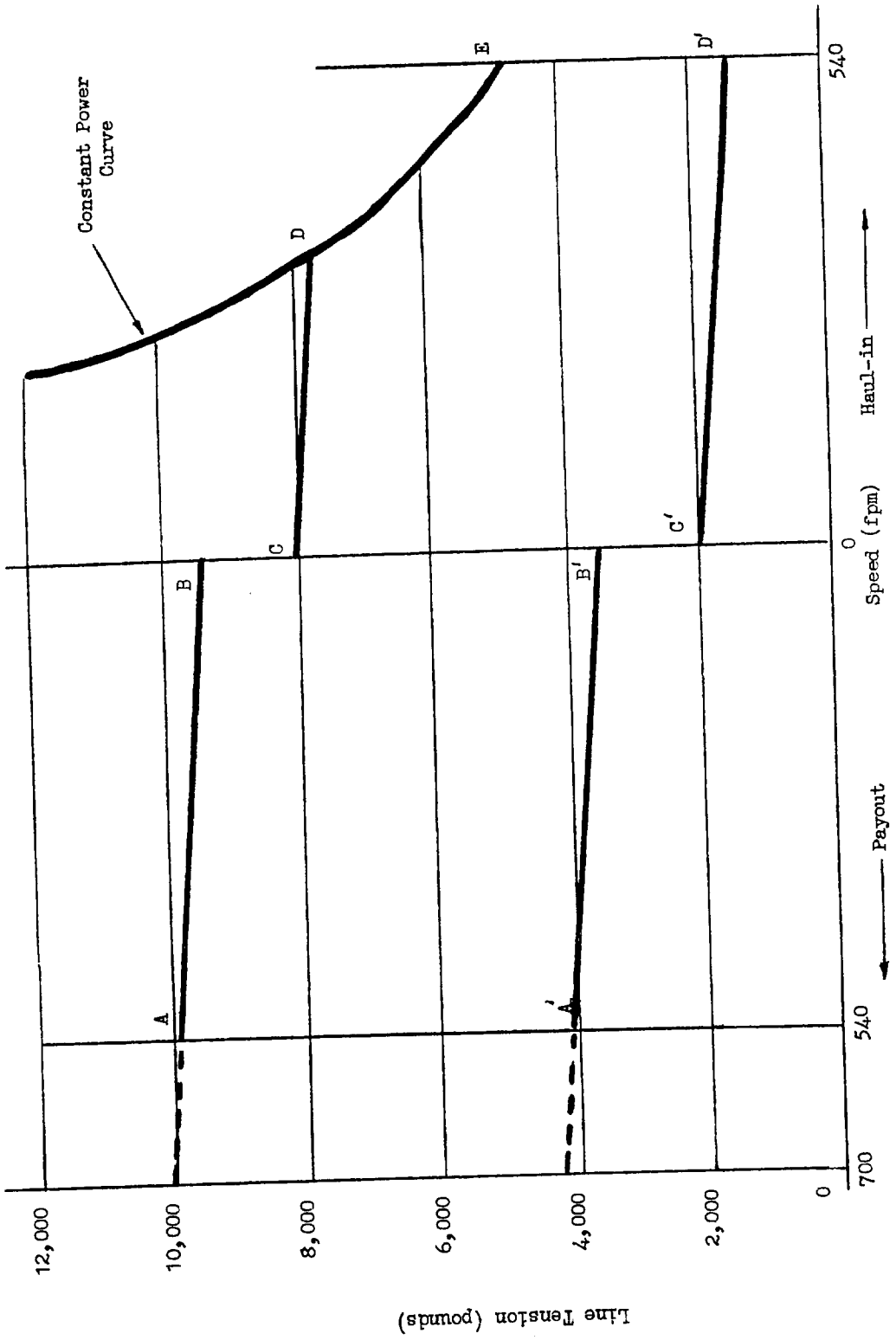



Figure 2 - Speed tension curves.

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